Case study on water pollution in nashik

Submitted by: Yuvraj Vinayak Suryawanshi
Class/Roll No.: 181
Subject: Environmental Studies
Academic Year: 2026-2027

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1.Introduction

Water pollution occurs when harmful substances such as chemicals, sewage, plastics and other waste materials enter water bodies and affect humans, plants, animals and aquatic ecosystems.

Urbanisation, domestic wastewater, agricultural activities, industrial discharge and improper waste disposal are major sources of water pollution.

Nashik is a rapidly developing city where these activities can contribute to water-quality problems. This report studies the major causes, effects and control measures of water pollution in Nashik.

Objectives of the Study

- To understand water pollution and its importance.
- To identify major sources of water pollution in Nashik.
- To study its effects on health and the environment.
- To analyse major local causes of water pollution.
- To suggest effective control measures.
- To understand the role of citizens and authorities

2.Overviewof water Pollution

Water pollution refers to the contamination of water bodies by harmful substances that can affect living organisms and environmental quality.

The pollutants may enter water naturally or through human activities. However, in cities, human activities are generally the major contributors to water pollution.

Major Water Pollutants:

Water pollutants are harmful substances that affect water quality, human health and aquatic ecosystems.

- Sewage: Domestic wastewater containing organic matter and harmful microorganisms.
- Industrial Chemicals: Released from factories and industrial processes.
- Fertilisers & Pesticides: Enter water bodies through agricultural runoff.
- Plastics: Waste materials that pollute rivers, lakes and other water sources.

Types of Water Pollution

Water pollution can mainly be classified into surface water and groundwater pollution. Surface water pollution occurs due to sewage discharge, industrial waste, agricultural runoff and plastic

waste, while groundwater pollution is caused by chemical leakage, excessive fertiliser use, landfill waste and contaminated water entering the soil.

3.Water Pollution In Nashik

Nashik has experienced considerable urban and infrastructural development. As the city expands, the amount of domestic wastewater, commercial activities, industrial discharge and solid waste also increases.

Water pollution is not necessarily uniform throughout the city. Areas near sewage discharge points, industrial zones, agricultural fields or polluted water bodies may experience greater local water-quality problems.

Important Local Sources

Water pollution in Nashik can mainly arise from untreated sewage, industrial effluents, agricultural runoff, improper waste disposal, plastic waste and leakage from drainage systems. Rainfall and seasonal changes can also influence how pollutants enter and spread through water bodies**.**

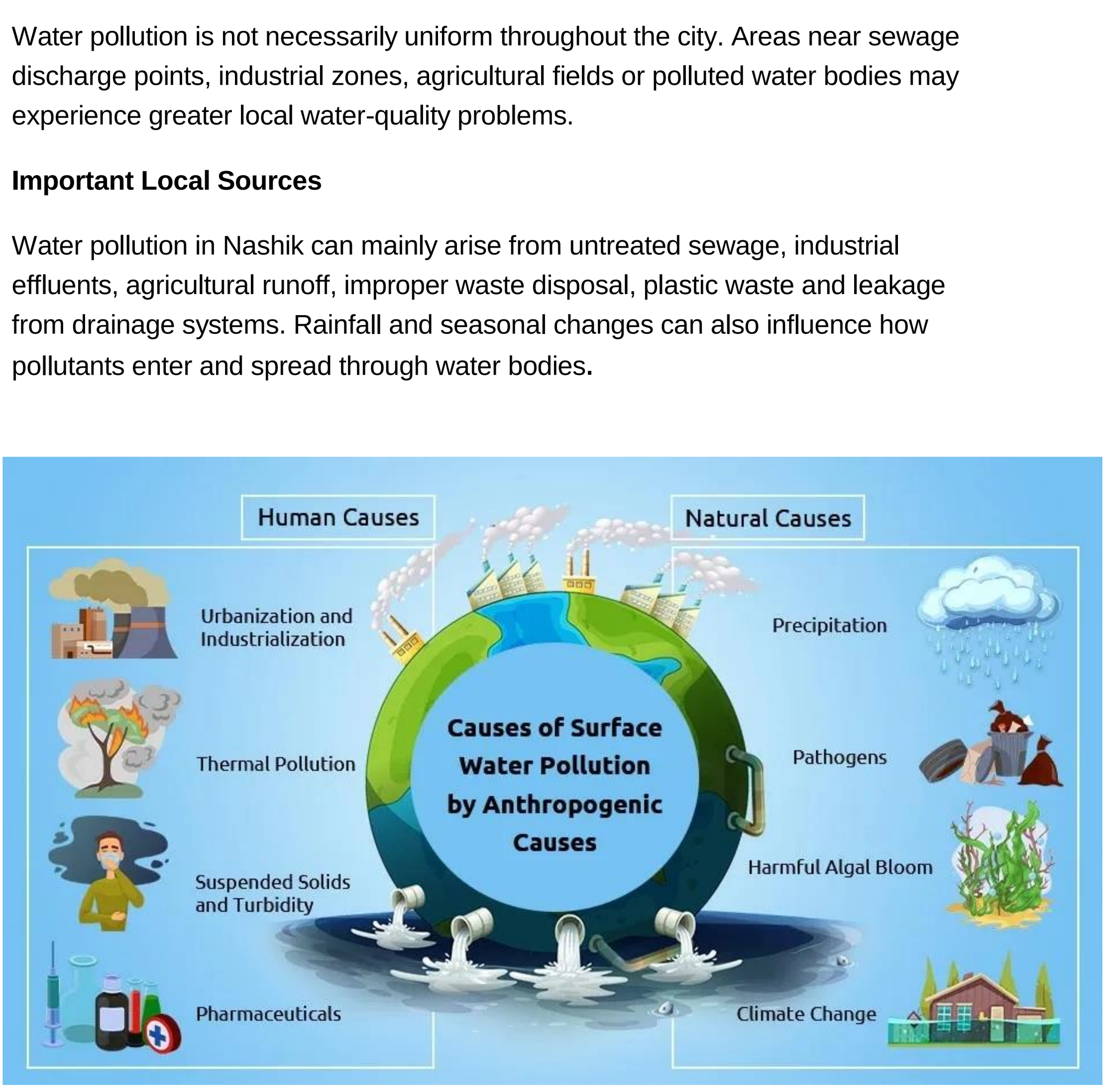


Fig. 1: Major Sources of water Pollution in Nashik City

4. Major Causes of Water Pollution

1) Domestic Sewage and Wastewater

Domestic sewage is a major source of water pollution in urban areas. In Nashik, increasing population and residential development can increase the quantity of wastewater generated from homes, apartments and commercial areas. Untreated sewage may contain organic matter, detergents, nutrients and harmful microorganisms that enter nearby water bodies. Improper drainage and leakage from sewage systems can further increase water pollution. Areas without adequate treatment facilities may experience greater local water-quality problems. Disposal of household waste and wastewater into drains can also introduce pollutants into rivers and streams.

Major factors contributing to domestic water pollution include:

- Increasing population and wastewater generation.
- Discharge of untreated sewage.
- Poor drainage and sewage systems.
- Improper disposal of household waste.
- Leakage from sewage pipelines.

Domestic wastewater can introduce pollutants such as organic matter, nutrients, detergents, bacteria and other harmful substances. Long-term exposure to polluted water can contribute to waterborne diseases and deterioration of aquatic ecosystems.

Possible control measures: Proper sewage treatment, improved drainage systems, regular maintenance of pipelines and responsible disposal of household wastewater can help reduce domestic water pollution.

2) Industrial Activities

Rapid industrial development in Nashik can generate significant amounts of wastewater. Activities such as manufacturing, chemical processing and cleaning operations may produce liquid waste containing harmful substances. Improperly treated industrial effluents can enter nearby rivers, streams and groundwater if proper control measures are not followed.

3) Agricultural Runoff

Agricultural runoff is another important source of water pollution.

Fertilisers, pesticides and other agricultural chemicals can be washed from fields into nearby water bodies during rainfall or irrigation.

Contributing factors:

- Excessive use of fertilisers.
- Improper use of pesticides.
- Runoff from agricultural fields.

4) Industrial Wastewater

Industrial activities can release chemicals, oils and other pollutants through manufacturing

processes, cleaning operations and wastewater discharge. Improper treatment of industrial effluents can negatively affect water quality and aquatic life. Proper wastewater treatment systems and regular monitoring are important to reduce industrial water pollution.
Major sources include:

- Industrial wastewater discharge.
- Chemicals and oils from industries.
- Leakage or improper disposal of industrial waste.

5) Open Disposal of Waste

Dumping plastic, paper, food waste and other materials in open areas can eventually pollute nearby water bodies. During rainfall, waste and harmful substances may be carried into rivers, streams and drainage systems. Improper waste disposal and lack of awareness are common reasons for this practice. Avoiding open dumping and following proper waste management can help reduce water pollution.

6) Leakage and Improper Wastewater Disposal

Leakage from sewage pipelines, septic systems and drainage networks can allow contaminated wastewater to enter soil and groundwater. Such wastewater may contain harmful microorganisms, chemicals and organic pollutants. Reducing unnecessary discharge and maintaining sewage systems properly can help prevent groundwater contamination and improve local water quality.

5.Effects of water Pollution

Water pollution can have harmful effects on human health, aquatic organisms, plants and the overall environment. The severity of these effects depends on the type and concentration of pollutants, as well as the duration of exposure. In urban areas, continuous contact with polluted water can reduce the quality of life and create both short-term and long-term health concerns.

1) Effects on Human Health

Water pollution directly affects human health because contaminated water may contain harmful microorganisms, chemicals and toxic substances. Consumption or prolonged contact with polluted water can cause stomach infections, skin problems and other waterborne diseases. Contaminated water may also affect people when it is used for drinking, cooking or bathing.

Common effects of water pollution include:

- Stomach infections and waterborne diseases.
- Skin irritation and infections.
- Diarrhoea and other digestive problems.
- Health problems due to harmful chemicals.

Children, elderly people and individuals using contaminated water sources may be more vulnerable to its effects. Therefore, maintaining clean and safe water is essential for

protecting public health and improving the overall quality of urban life.

2) Effects on the Environment

Water pollution can damage the natural environment by affecting rivers, lakes, groundwater, soil, vegetation and aquatic ecosystems. Pollutants can reduce water quality, disturb aquatic habitats and affect the natural balance of ecosystems.

Major effects include:

• Damage to aquatic habitats and ecosystems.

• Pollution of groundwater and soil.

• Reduction in water quality and aquatic life.

Therefore, controlling water pollution is important for protecting natural water resources and maintaining ecological balance.

3) Effects on Aquatic Plants

Aquatic plants are sensitive to water pollution because excessive nutrients, chemicals and other pollutants can interfere with their normal growth and development. Long-term exposure to polluted water can damage plants and affect aquatic ecosystems.

Water pollution can negatively affect aquatic plants by reducing water quality and limiting sunlight penetration. Excessive nutrients may cause unwanted algal growth, reduce dissolved oxygen and disturb normal plant growth. Prolonged exposure to pollutants may reduce plant productivity and affect the overall health of aquatic ecosystems.



Fig2.Effect On Plants

Water pollution in Nashik is caused by several activities such as domestic sewage discharge, industrial wastewater, agricultural runoff, improper waste disposal and drainage leakage. These sources introduce pollutants like organic matter, harmful microorganisms, chemicals, fertilisers, pesticides and plastics into water bodies.
The effects of these pollutants may include waterborne diseases, skin irritation, reduced water quality, damage to aquatic life, groundwater contamination and environmental degradation. Since pollution comes from multiple sources, effective control requires a combined approach involving citizens, industries and local authorities.
Source → Pollutant → Impact
• **Domestic sewage → Organic waste, bacteria, nutrients → Waterborne diseases**

• **Industrial wastewater → Chemicals, oils, toxic substances → Health/environmental effects**
• **Agricultural runoff → Fertilisers, pesticides → Water-quality deterioration**
• **Improper waste disposal → Plastics and solid waste → Aquatic pollution**
• **Drainage leakage → Sewage, microorganisms, chemicals → Groundwater contamination**
• **Open dumping → Organic waste, plastics → Pollution of water bodies**

Key Observation

Water pollution in a city is usually not caused by a single source. Multiple sources operate simultaneously and together affect surface water, groundwater and the overall aquatic environment.

7. Possible Control Measures

Control of Domestic Sewage

Domestic sewage is a major source of water pollution in urban areas. Proper collection, treatment and disposal of wastewater can prevent harmful substances and microorganisms from entering rivers, lakes and groundwater.

Important measures include:

• Treat domestic sewage before releasing it into water bodies.
• Maintain proper drainage and sewage systems.
• Prevent leakage from sewage pipelines.
• Avoid dumping household waste into drains and water bodies.
• Reuse treated wastewater wherever appropriate.

These measures can reduce water pollution and help maintain cleaner water resources in Nashik.

Control of Industrial Wastewater

Industrial activities can generate wastewater containing chemicals, oils and other harmful substances. Proper treatment of industrial effluents can prevent pollutants from entering nearby rivers, streams and groundwater.

Important measures include:

• Treat industrial wastewater before discharge.
• Follow proper effluent management practices.
• Prevent leakage and accidental discharge of chemicals.

These practices can significantly reduce industrial water pollution.

Control of Agricultural Runoff

Agricultural runoff is an important source of water pollution. Fertilisers and pesticides can be carried into nearby water bodies during rainfall. Proper agricultural practices can help reduce the amount of pollutants entering water resources.

Important control measures include:

• Use fertilisers and pesticides in appropriate quantities.
• Avoid excessive use of chemical fertilisers.
• Maintain vegetation or buffer zones near water bodies.
• Prevent agricultural waste from entering drains and rivers.
• Promote environmentally friendly farming practices.

These measures can help reduce agricultural pollution and improve overall water quality.

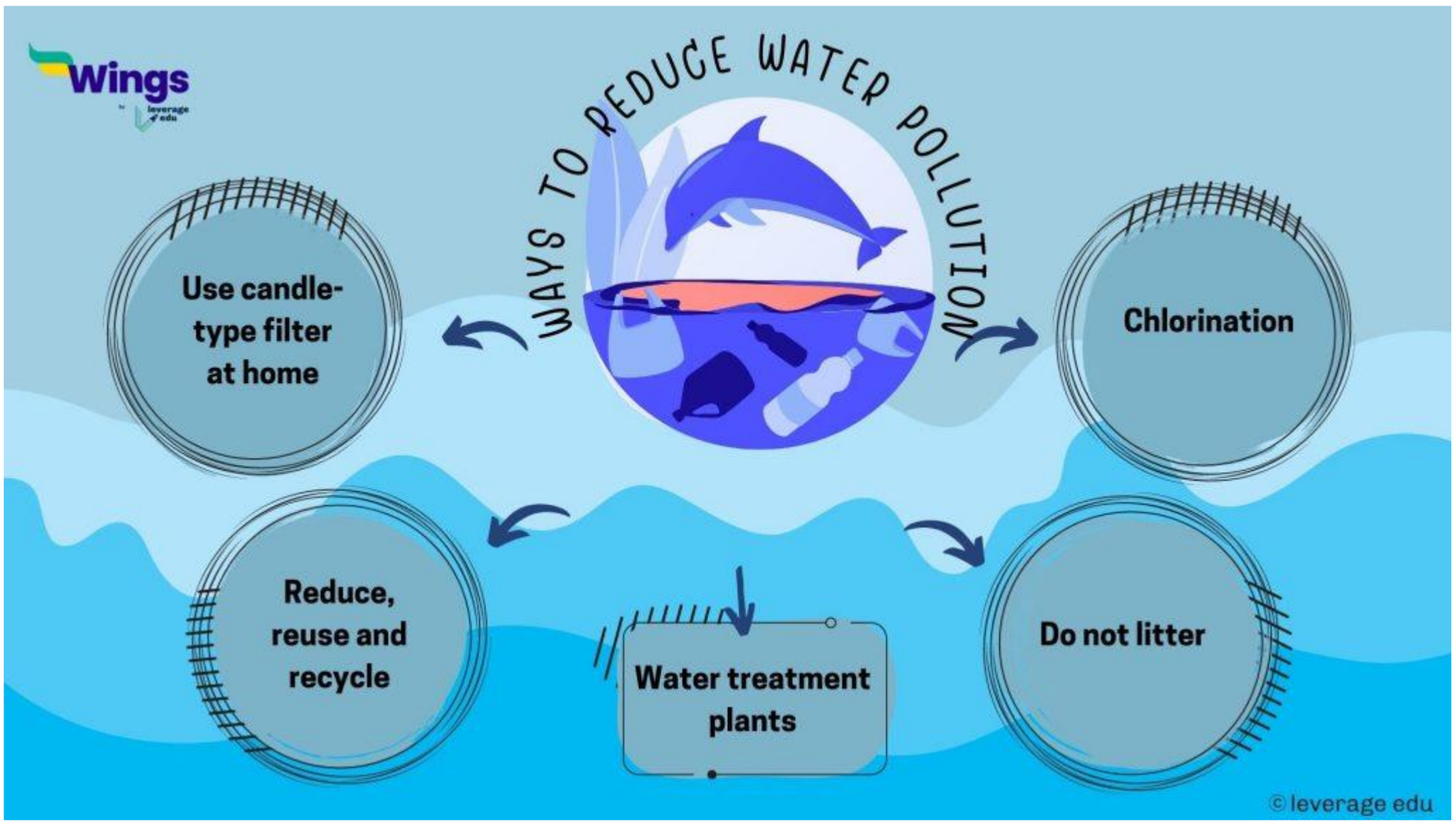


Fig3. Reduce Pollution Using Sustainable Energy

These measures help reduce water pollution and improve overall water quality.

Control of Industrial Pollution:

Industrial water pollution can be reduced by using proper wastewater treatment systems, cleaner production technologies and pollution-control methods. Regular maintenance and monitoring of industrial effluents can further reduce harmful pollutants. Following environmental regulations and improving water-use efficiency can help industries maintain better water quality in and around Nashik.

Prevention of Improper Waste Disposal:

Improper disposal of waste can introduce plastics, organic waste and harmful substances into water bodies. It can be prevented through proper waste segregation, collection and disposal. Organic waste can be composted, while recyclable materials should be recycled. Public awareness can also help discourage dumping of waste near rivers, lakes and drains and reduce water pollution.

8. Role of Citizens & Role of Local Authorities:

Role of Citizens

Citizens have an important role in controlling water pollution.

Individual Actions:

• Avoid dumping waste into rivers, lakes and drains.

• Dispose of household waste properly.

• Reduce the use of single-use plastics.

• Avoid releasing harmful chemicals into drains.

• Save water and prevent unnecessary wastage.

• Keep surroundings and water sources clean.

Community Actions:

• Organise cleanliness drives near water bodies.

• Conduct awareness campaigns.

• Promote proper waste segregation.

• Support local water-pollution control initiatives.

Role of Local Authorities

Local authorities play an important role in controlling water pollution by monitoring water quality and enforcing environmental regulations. They can improve sewage treatment, maintain drainage systems, control industrial wastewater discharge, manage solid waste and prevent dumping near water bodies. Developing clean public spaces and conducting awareness programmes can further support pollution reduction. Cooperation between authorities, industries and citizens is essential for long-term improvement in water quality.

9. Awareness and Sustainable Solutions

Water pollution cannot be solved only through government action. Sustainable solutions require cooperation between society, industries and authorities. Responsible water use, proper wastewater management, waste reduction and public participation can help protect water resources for future generations.



Fig5.Recycle-Reuse-Reduce.

The focus should be on preventing water pollution at its source rather than dealing with its effects after contamination has already occurred.

9.Conclusion

Water pollution is a serious environmental issue that can affect human health, aquatic life and the overall quality of the environment. In Nashik, major sources such as domestic sewage, industrial wastewater, agricultural runoff, improper waste disposal and drainage leakage contribute to water-quality deterioration.

The analysis shows that controlling water pollution requires proper sewage treatment, effective wastewater management, regular drainage maintenance, industrial effluent control and responsible waste disposal. Public awareness and active participation of citizens are also important for preventing pollution.

Therefore, water pollution control is a shared responsibility of citizens, industries and local authorities. With proper planning, monitoring and sustainable practices, Nashik can move towards cleaner water resources and a healthier environment.



Fig6. Comparison Image of Best and Worst City.

10.References

1. Central Pollution Control Board (CPCB) – Water Quality Monitoring and Pollution Control Reports.

2. Maharashtra Pollution Control Board (MPCB) – Water Quality and Environmental Reports.

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